

Introduction to hierarchical Bayesian modeling

Instructor: Kiona Ogle
Kiona.Ogle@gmail.com

Lecture content

- Simple example with Binomial tree mortality data
 - Non-hierarchical vs hierarchical versions
 - Borrowing of strength
 - Model complexity & effective # of parameters
- Detailed example with photosynthesis light-response curves
 - Multi-level, non-linear hierarchical model
 - Reparameterization considerations
 - Linkages to “fixed” and “random” effects concepts
 - Code and example results
- Appendix
 - Other reparameterization options
 - Additional considerations
 - *To be posted on GitHub*

Recall: Simple example

Objective: estimate the probability of a tree dying during a drought (θ)

Data: total number of trees that died (Y) out of N total trees in the plot;
(i.e., $N-Y$ survived); surveyed P randomly located plots.

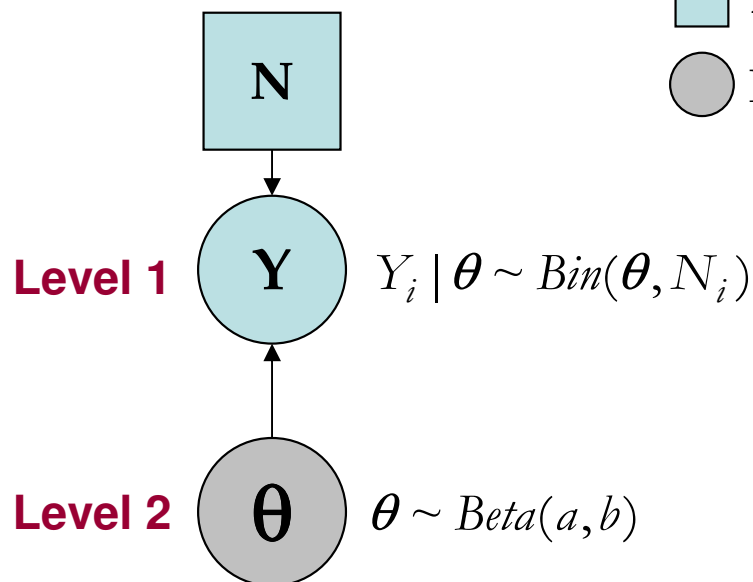
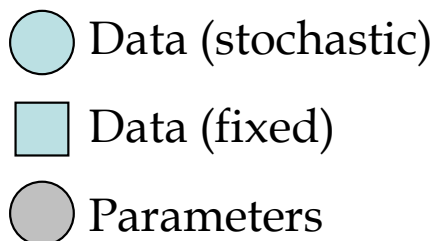
Model: simple Binomial likelihood for number of dead trees, simple conjugate beta prior for scalar parameter, θ .

Potential consideration: should each plot have its own probability of dying? That is, should θ vary by plot?

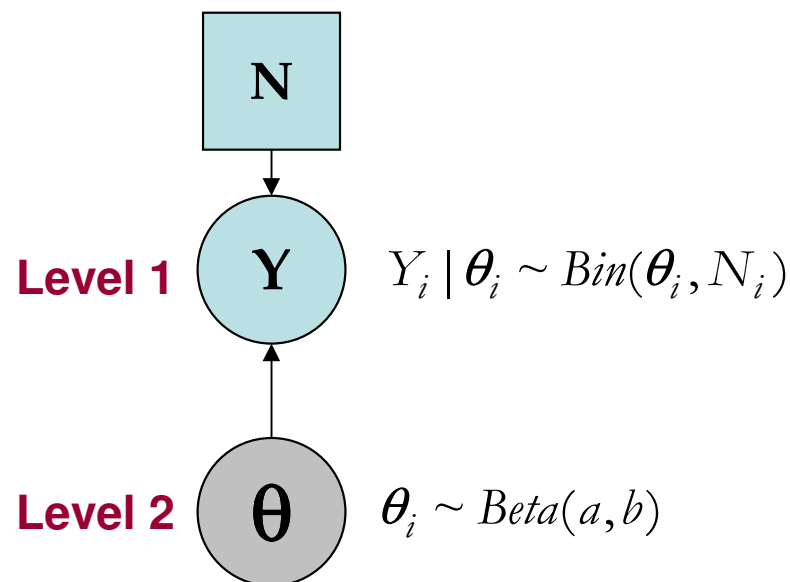
Plot-level parameters: treat the sampled plots as a random sample from a potential population of plots within the focal woodland. Suggests a **hierarchical** model (prior) for plot-level parameters.

Plot-level parameters

Original
(common θ)



Non-hierarchical,
plot-level parameters
(i.e., plot-specific θ 's)



posterior $\propto$ likelihood $\times$ prior

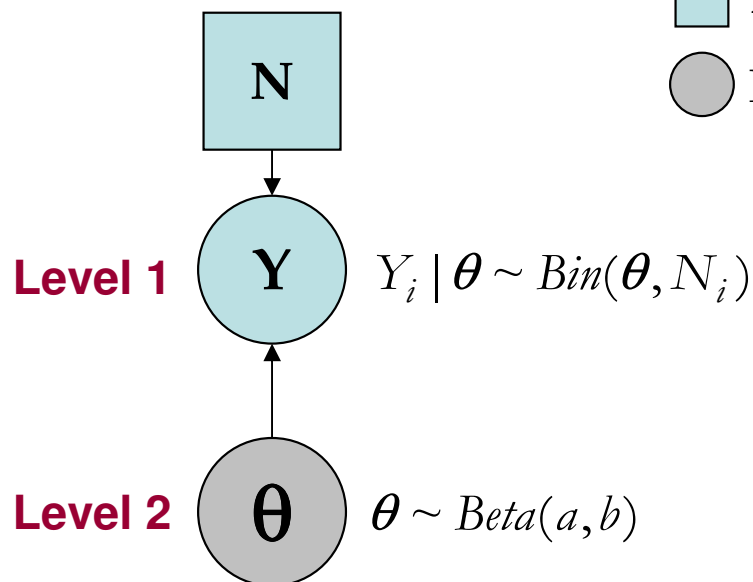
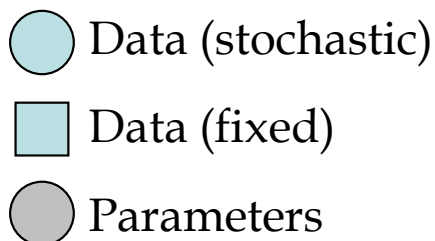
$$[\theta | Y] \propto [Y | \theta][\theta]$$

$$[\theta | Y] \propto ??$$

$$[\theta | Y] \propto [Y | \theta][\theta]$$

Hierarchical version

Original
(common θ)

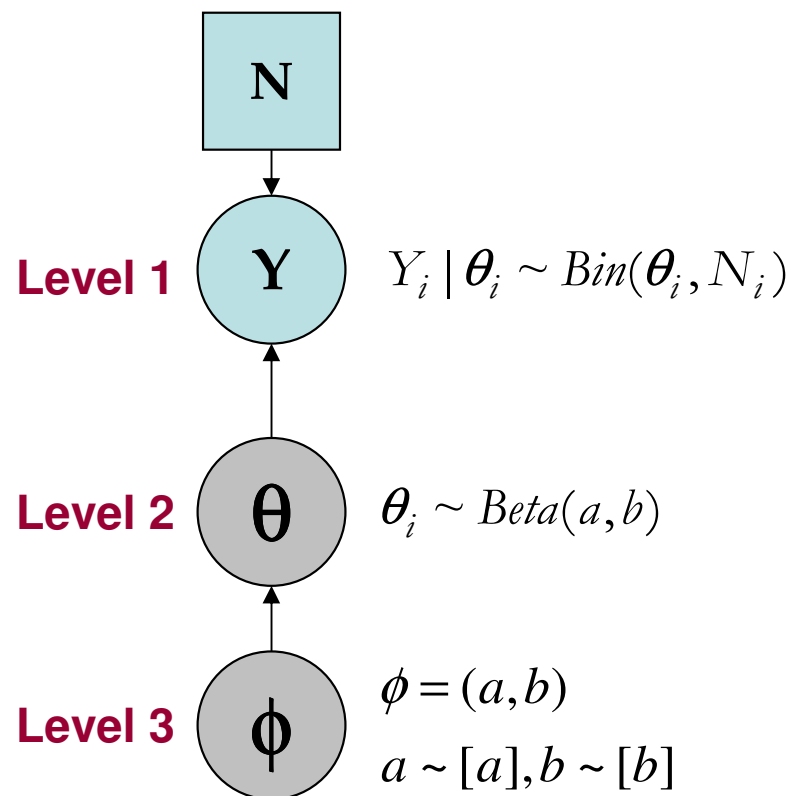


$\text{posterior} \propto \text{likelihood} \times \text{prior}$

$$[\theta | Y] \propto [Y | \theta][\theta]$$

$$[\theta, \phi | Y] \propto ??$$

**Hierarchical
parameter model**
(e.g., plot-specific θ 's)



$$[\theta, \phi | Y] \propto [Y | \theta][\theta | \phi][\phi]$$

$\text{posterior} \propto \text{likelihood} \times \text{prior} \times \text{hyperprior}$

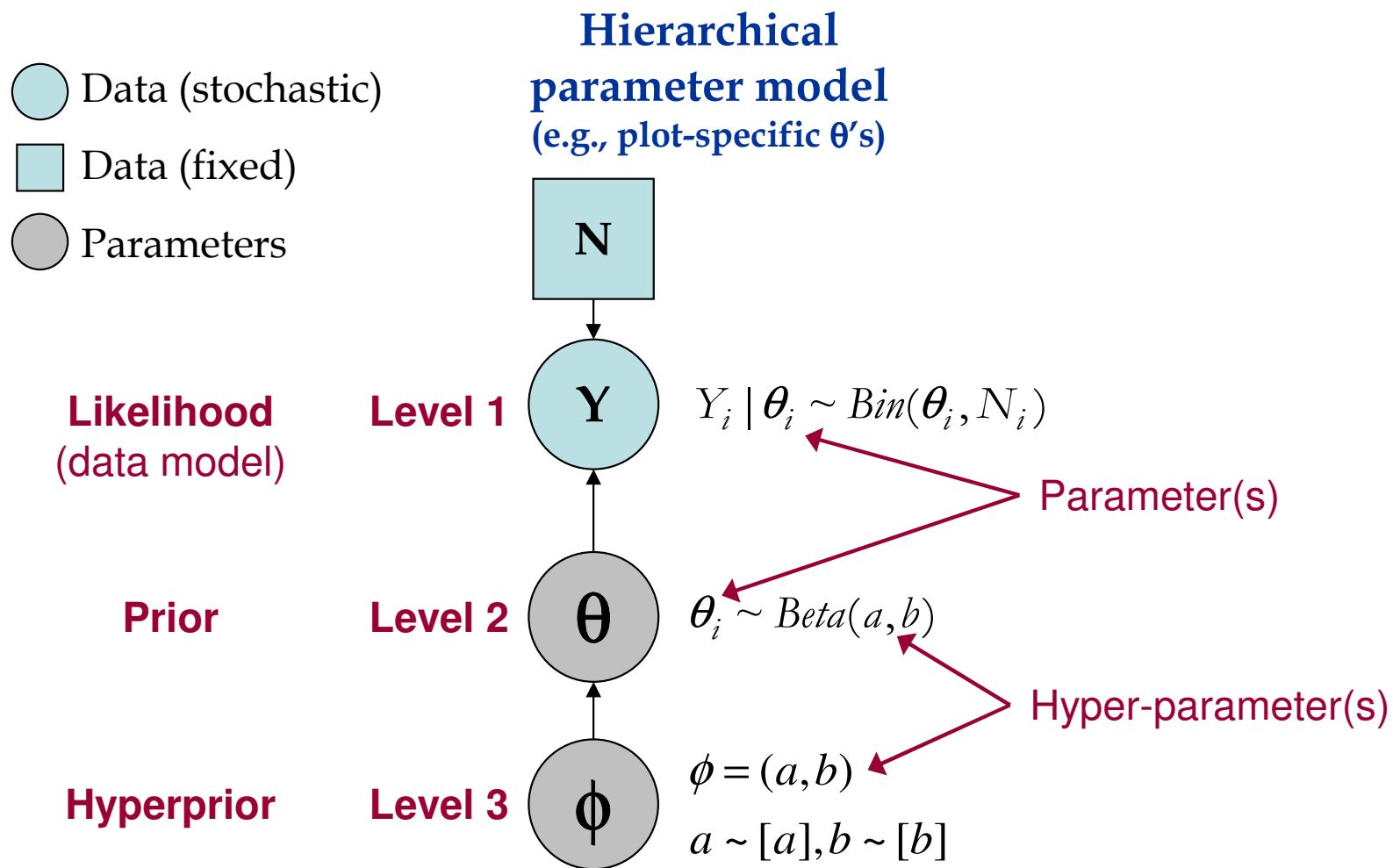
Motivation for hierarchical models

- Represent variation among individuals, plots, times, etc.
- Representation of processes occurring at multiple scales:
 - Spatial scales (e.g., plots, watersheds, basins, regions, etc.)
 - Temporal scales (e.g., hour, day, season, year, etc.)
 - Biological/ecological scales (e.g., cells, tissues, individuals, populations, species, communities, etc.)
- Accommodate different sources of uncertainty
 - E.g., separation of observation and process error
- Representation of latent processes
 - E.g., true, hidden, or unobservable processes

Motivation for hierarchical models

- Ability to factor complicated models into easy to “think about” conditional pieces
 - Think globally, model locally
 - Graphical models help with model building
- Accommodate experimental / sampling design
 - Often have hierarchical or nested structure (e.g., individuals within plots within sites)
- Facilitates integration of multiple data types
 - Of different spatial, temporal, and biological resolutions
 - Measured with varying precision

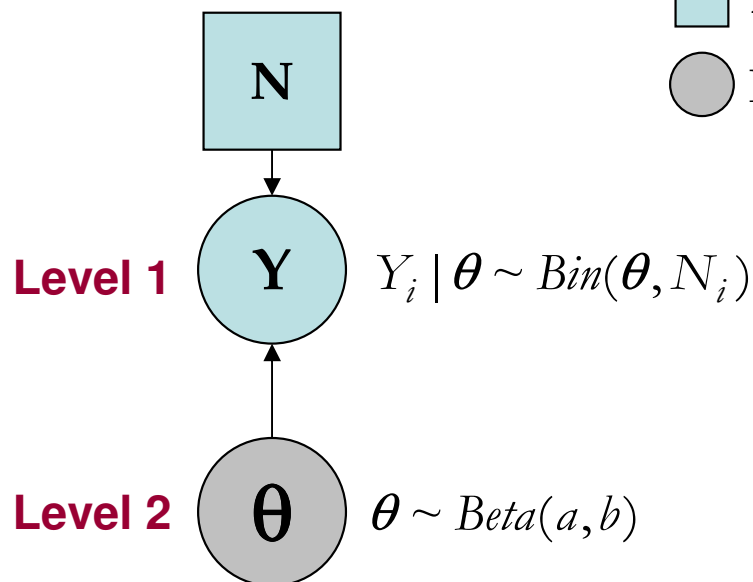
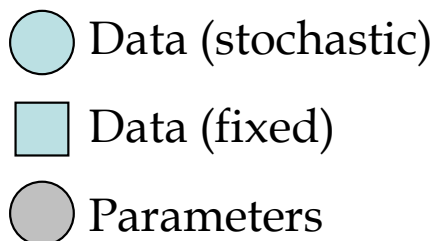
Return to: Hierarchical version



$$[\theta, \phi | Y] \propto [Y | \theta][\theta | \phi][\phi]$$

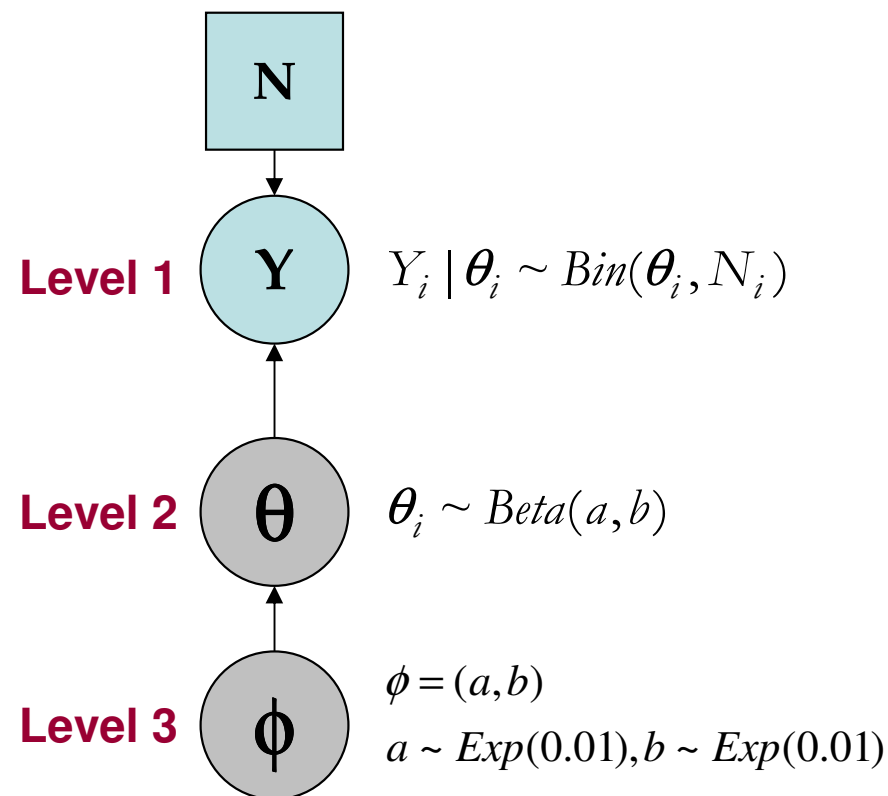
Statistical models

Original
(common θ)



$$[\theta | Y] \propto \theta^{\sum_{i=1}^P Y_i + a - 1} (1 - \theta)^{\sum_{i=1}^P (N_i - Y_i) + b - 1}$$

**Hierarchical
parameter model**
(e.g., plot-specific θ 's)



$$[\theta, a, b | Y] \propto \left(\prod_{i=1}^P \theta_i^{Y_i + a - 1} (1 - \theta_i)^{N_i - Y_i + b - 1} \right) e^{-0.01a} e^{-0.01b}$$

Example data

Plot	Y	N	X	Proportion
1	13	15	35.23	0.87
2	8	10	46.20	0.80
3	12	17	38.11	0.71
4	9	13	36.06	0.69
5	12	14	48.23	0.86
6	10	11	65.25	0.91
7	10	12	48.42	0.83
8	13	16	50.75	0.81
9	14	15	64.60	0.93
10	1	4	12.00	0.25

Y = number of dead trees

N = total number of trees

X = stand density

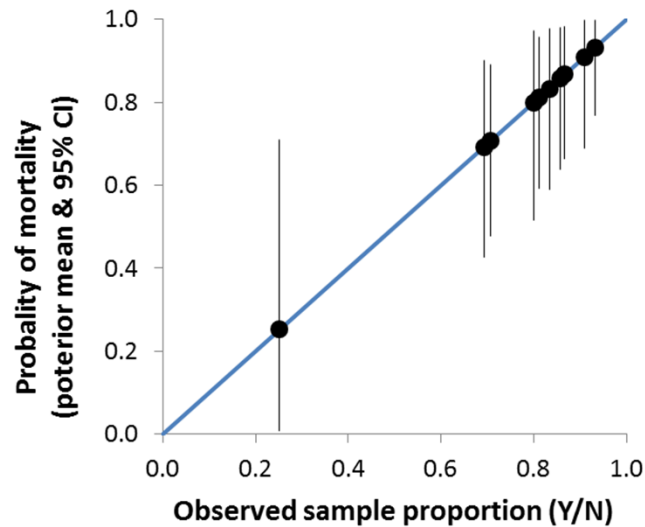
Proportion = proportion dead = Y/N

Borrowing of strength

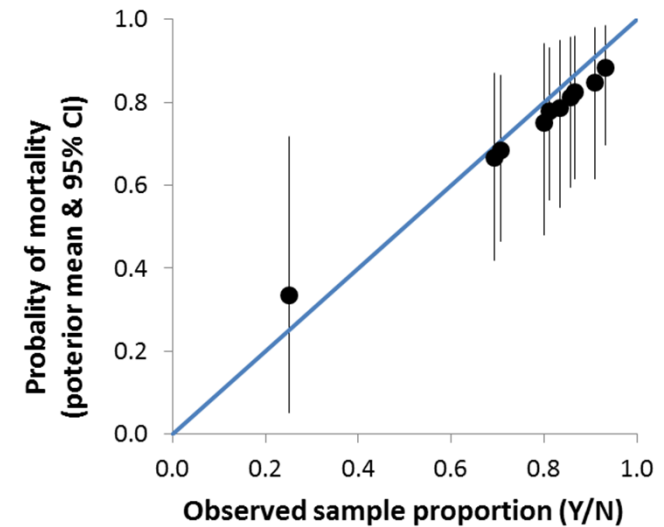
- Sharing of information among unknown quantities (e.g., plot-level θ 's) in a hierarchical model
- Useful for informing quantities associated with low replication (low sample sizes)
- Results in *shrinkage* towards population mean (e.g., individual θ 's pulled towards $E(\theta)$)
- Degree of shrinkage depends on sample sizes and within versus between group variability

Borrowing of strength

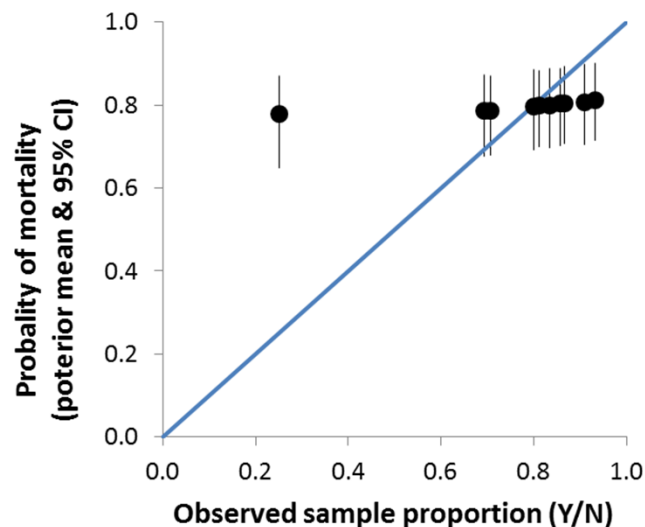
Non-hierarchical θ : Beta(0.01,0.01) priors



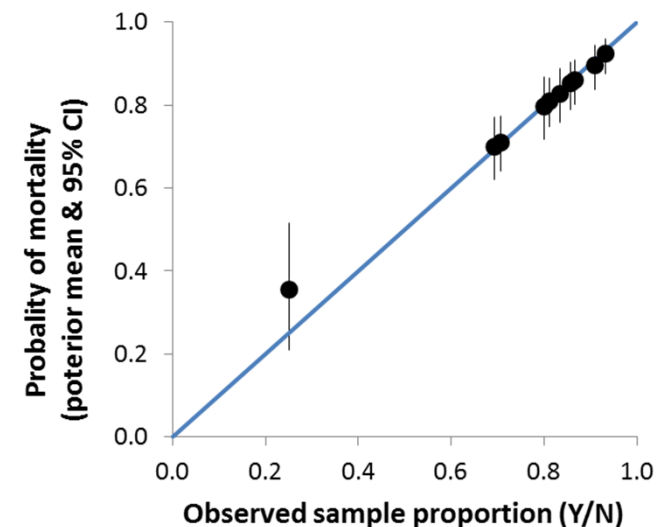
Non-hierarchical θ : U(0,1) priors



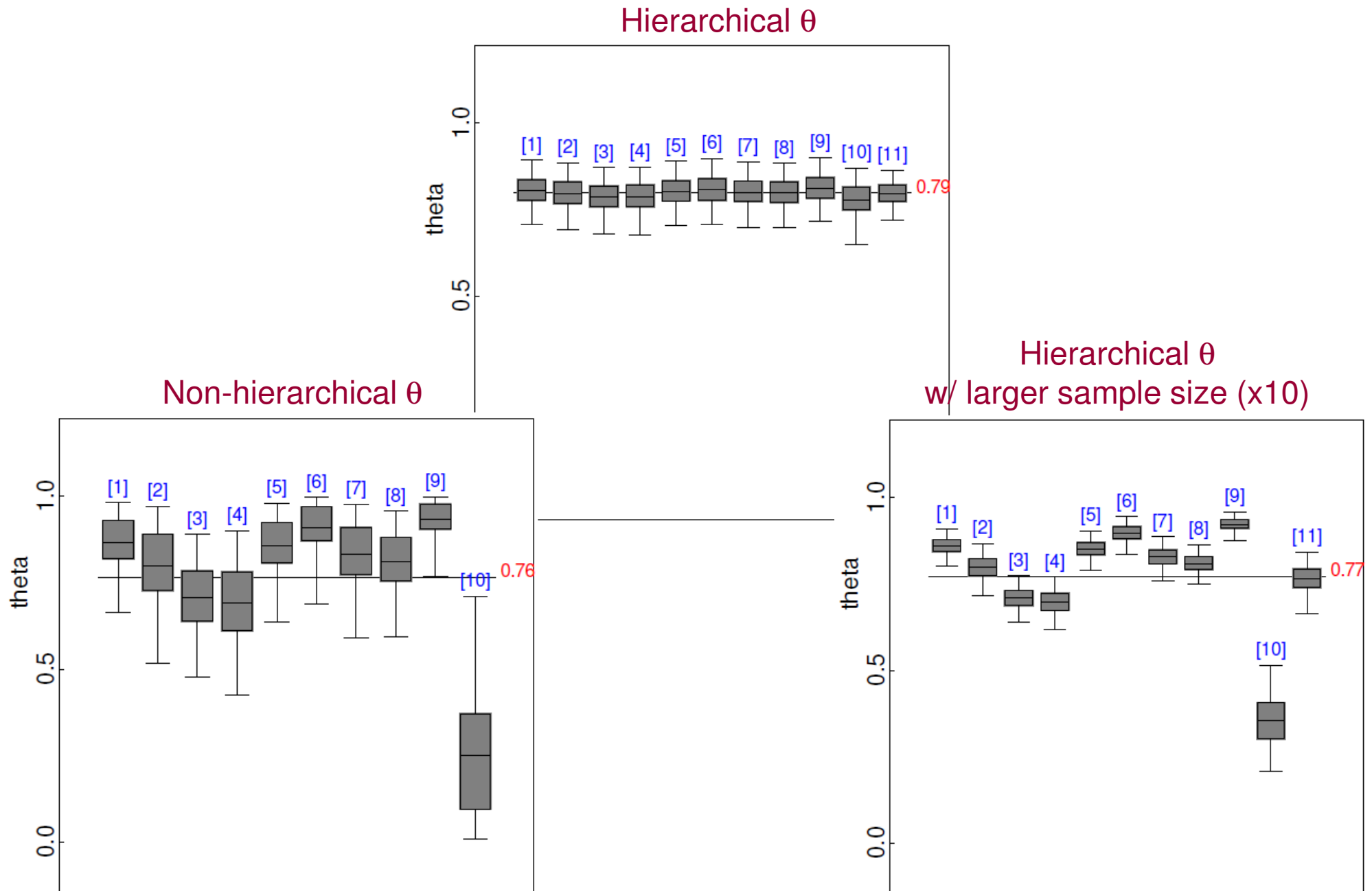
Hierarchical θ : original sample size



Hierarchical θ : larger sample size (x10)



Borrowing of strength



Effective # parameters

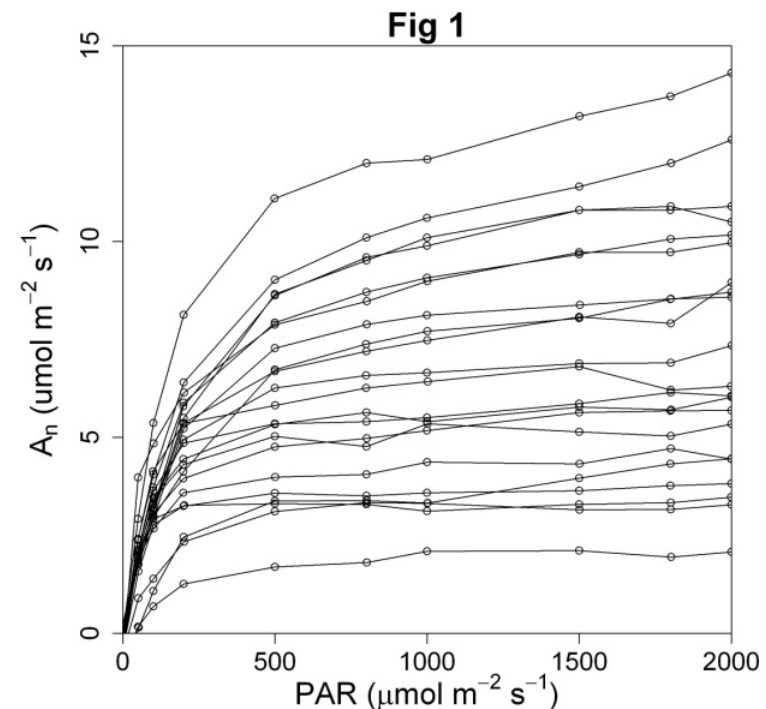
Model	Countable # parameters		Effective # parameters	
Common θ				
Non-hierarchical plot-level θ w/ beta priors		??	??	
Non-hierarchical plot-level θ w/ $U(0,1)$ priors				
Hierarchical plot-level θ				
Hierarchical plot-level θ w/ larger sample size (x10)				

Effective # parameters

Model	Countable # parameters	Effective # parameters
Common θ	1	1.0
Non-hierarchical plot-level θ w/ beta priors	10	10.9
Non-hierarchical plot-level θ w/ $U(0,1)$ priors	10	7.4
Hierarchical plot-level θ	12	1.8
Hierarchical plot-level θ w/ larger sample size (x10)	12	9.2

A detailed example: Photosynthesis light-response curves

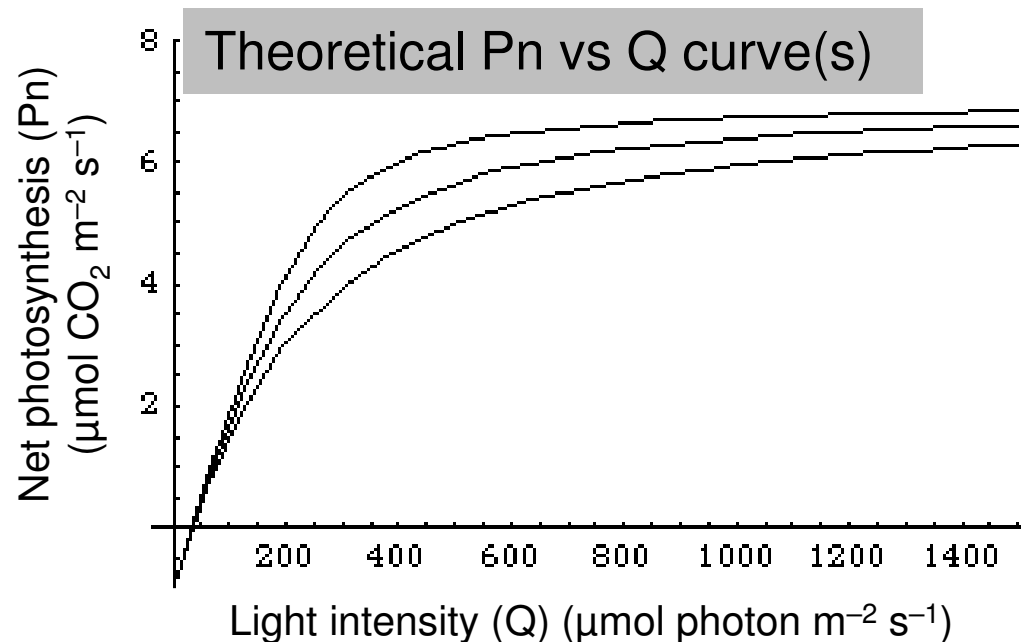
- Nonlinear relationship between photosynthesis and light intensity
- Features of light response curve may differ among leaves, plants, species, etc.
- Often measure multiple “curves” from a plant population of interest
- Each curve (e.g., leaf or plant) may be thought of as a random sample from a population of curves
- Fit non-linear response function to each curve to learn about population of interest



Photosynthesis light-response curves

- **Problem description**

- Wish to understand how net photosynthesis (P_n) responds to changes in light levels (Q).
- Determine if different species differ in their photosynthesis response to light
- Relationship between P_n and Q is known to follow a non-linear, asymptotic-type response:



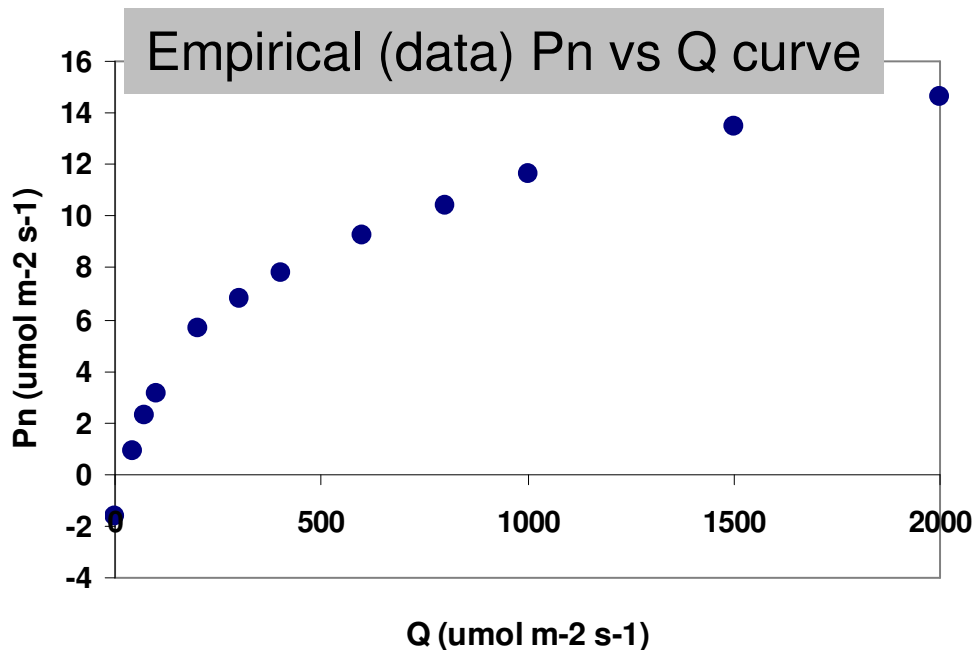
Photosynthesis light-response curves

- **Data collection**

- 8-10 observations of Pn vs Q were obtained per curve
- Total of 46 curves, for a total of 23 different plants, representing 4 different species

- **Likelihood**

- We may assume the following sampling distribution, which defines the likelihood of a single observation within a particular Pn vs Q curve:



$$Pn_i \sim \text{Normal}(\mu_i, \sigma^2)$$

precision $\tau = \sigma^{-2}$

Other potential distributions?

Non-rectangular hyperbola mean model

$$\mu_i = \frac{\alpha \cdot Q_i + P_{max} - \sqrt{(\alpha \cdot Q_i + P_{max})^2 - 4\alpha \cdot Q_i \cdot \beta \cdot P_{max}}}{2\beta} - R_d$$

Q_i = observed light level

Parameters of interest: P_{max} , R_d , α , β

P_{max} = max. gross photosynthesis rate

R_d = “dark” respiration rate

α = light-use efficiency

β = curvature parameter

Meaning of each parameter?

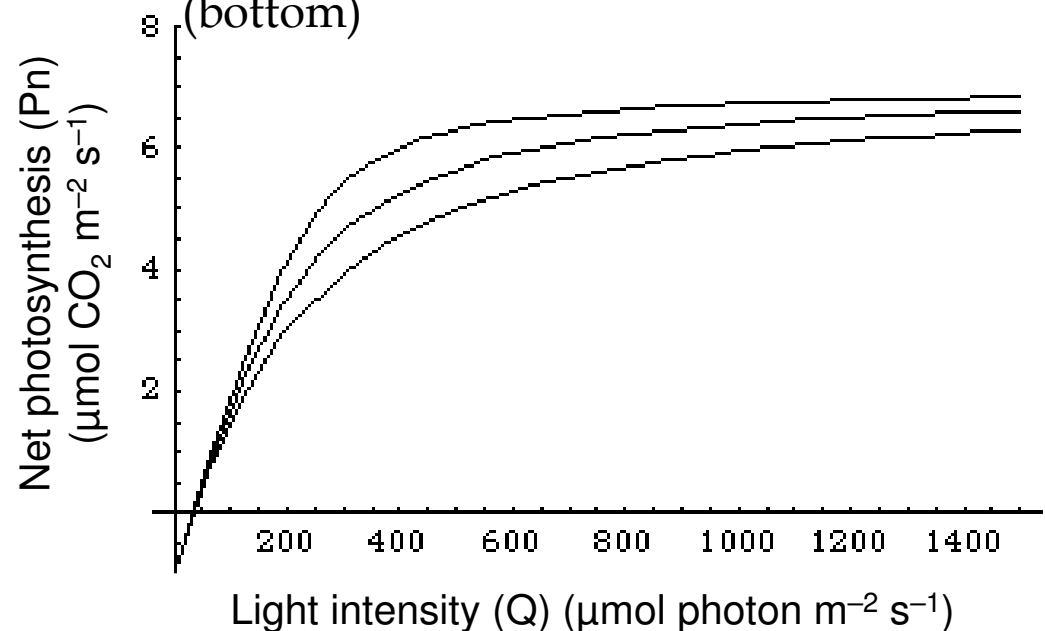
Support (domain) of each parameter?

These curves correspond to:

$P_{max} = 8$, $R_d = 1$, $\alpha = 0.03$, and

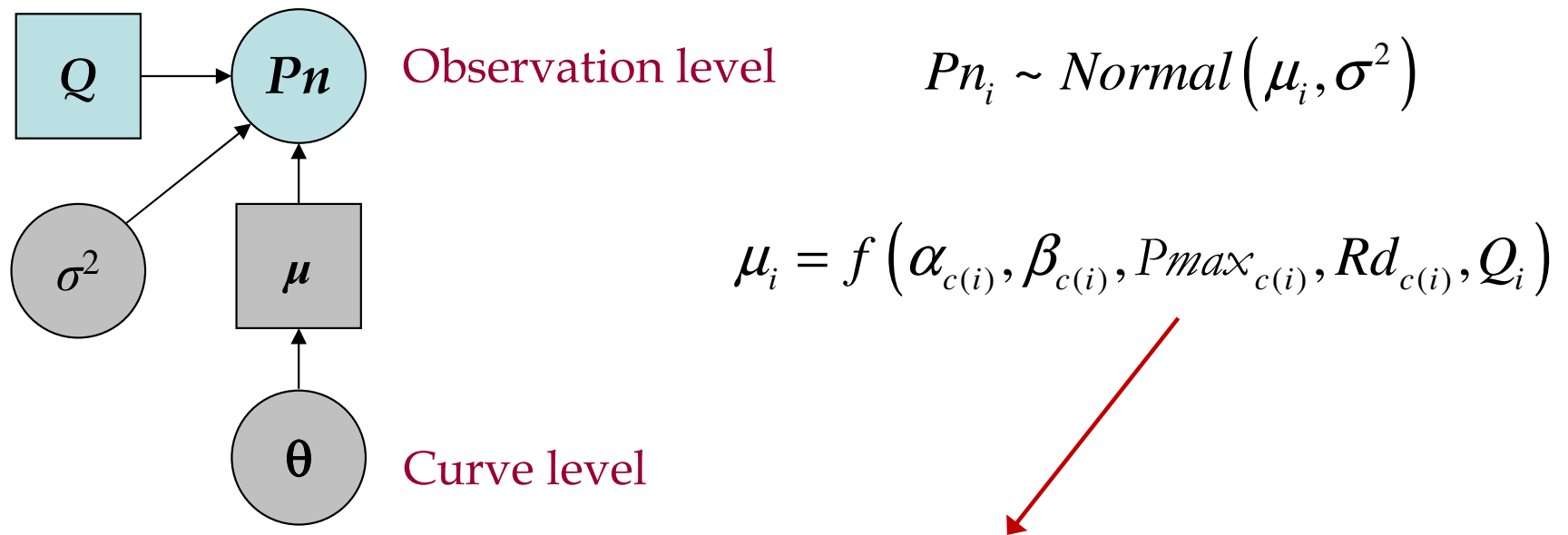
$\beta = 0.9$ (top), 0.75 (middle), and 0.5

(bottom)



Model specification

- For $i = 1, 2, \dots, N$ observations, $c = 1, 2, \dots, C$ curves, plant $p = 1, 2, \dots, P$ plants, and $s = 1, 2, \dots, S$ species:



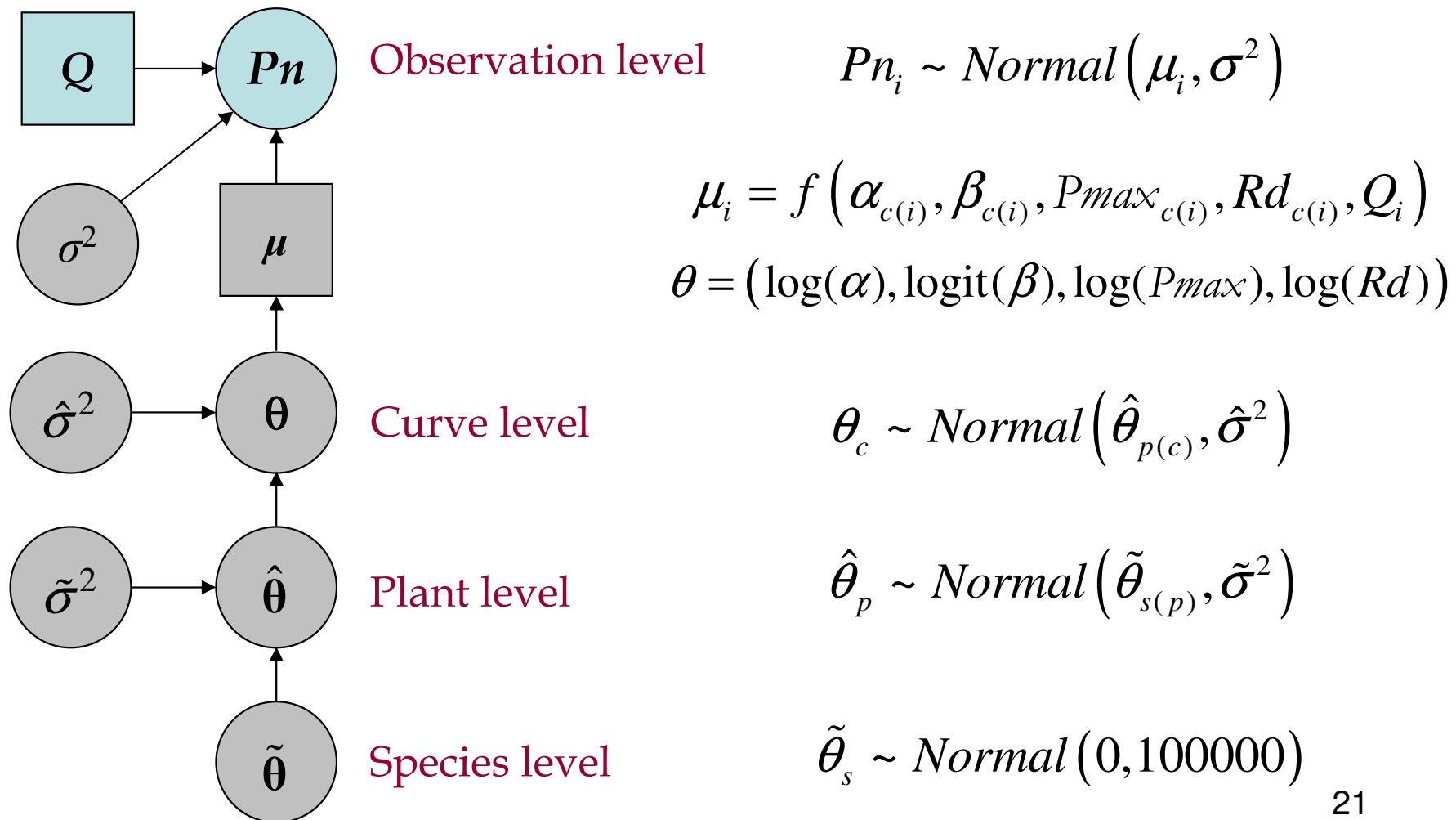
$$\mu_i = \frac{\alpha_{c(i)} \cdot Q_i + Pmax_{c(i)} - \sqrt{(\alpha_{c(i)} \cdot Q_i + Pmax_{c(i)})^2 - 4\alpha_{c(i)} \cdot Q_i \cdot \beta_{c(i)} \cdot Pmax_{c(i)}}}{2\beta_{c(i)}} - Rd_{c(i)}$$

- Reparameterize:** log- or logit-transform parameters, yielding θ . On transformed-scale, all θ 's have domain defined by real line:

$$\theta = (\log(\alpha), \text{logit}(\beta), \log(Pmax), \log(Rd))$$

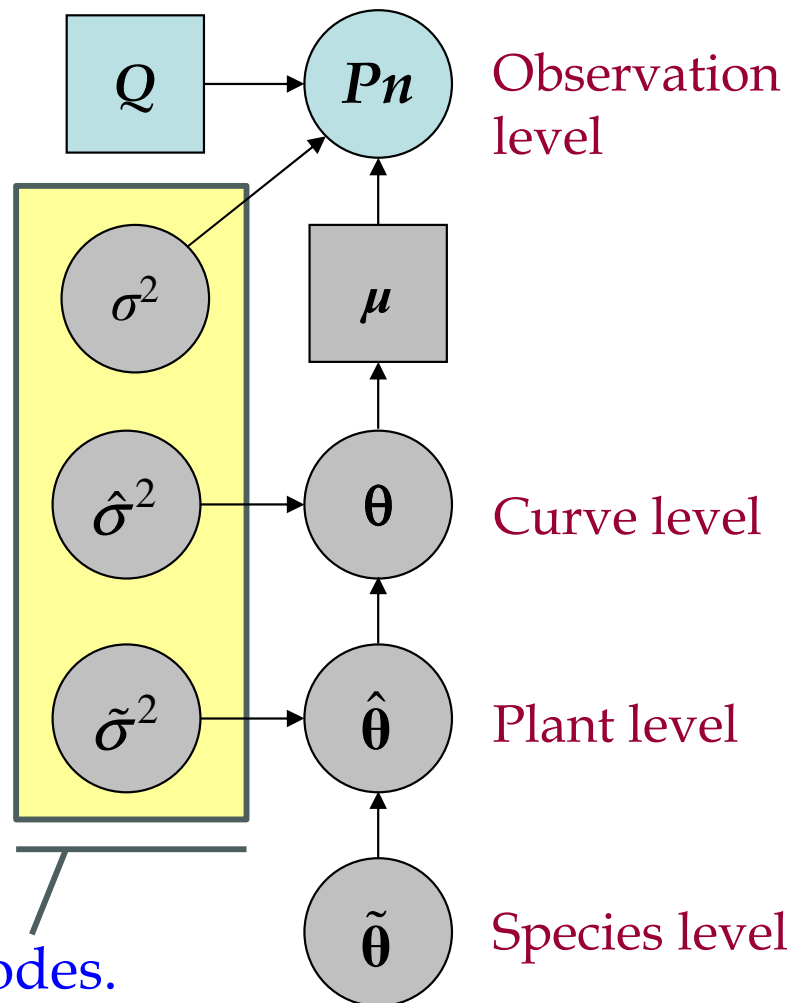
Hierarchical model specification

- For $i = 1, 2, \dots, N$ observations, $c = 1, 2, \dots, C$ curves, plant $p = 1, 2, \dots, P$ plants, and $s = 1, 2, \dots, S$ species:



Hierarchical model specification

- For $i = 1, 2, \dots, N$ observations, $c = 1, 2, \dots, C$ curves, plant $p = 1, 2, \dots, P$ plants, and $s = 1, 2, \dots, S$ species:



$$Pn_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = f(\alpha_{c(i)}, \beta_{c(i)}, Pmax_{c(i)}, Rd_{c(i)}, Q_i)$$

$$\theta = (\log(\alpha), \text{logit}(\beta), \log(Pmax), \log(Rd))$$

$$\theta_c \sim \text{Normal}(\hat{\theta}_{p(c)}, \hat{\sigma}^2)$$

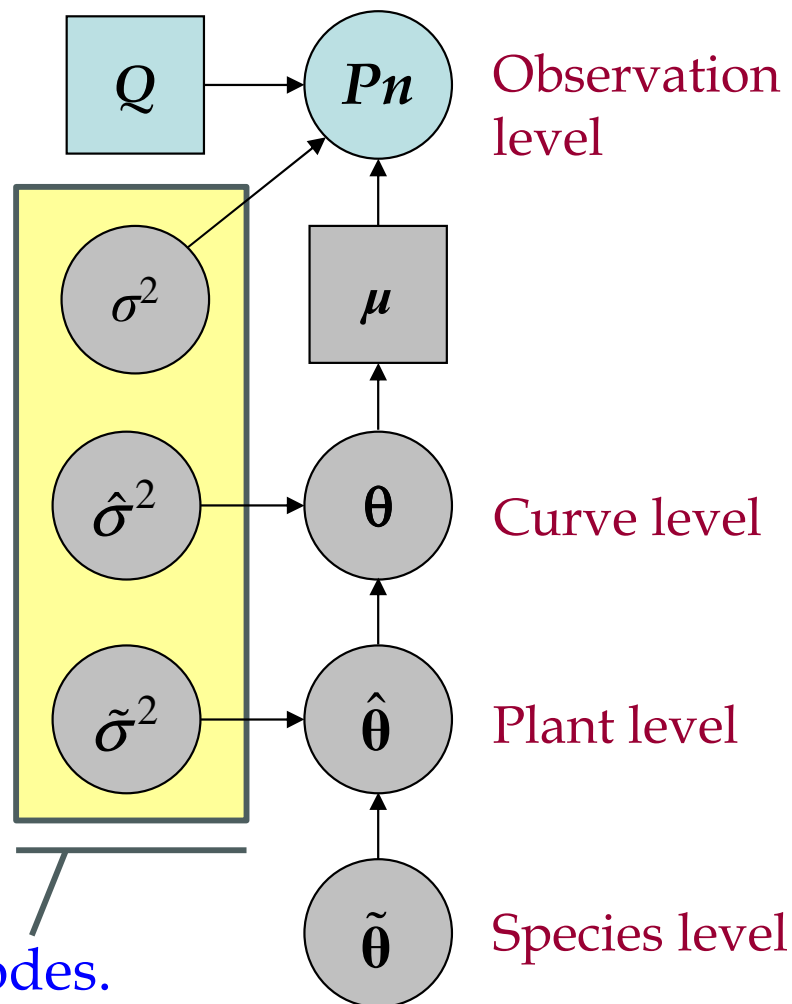
$$\hat{\theta}_p \sim \text{Normal}(\tilde{\theta}_{s(p)}, \tilde{\sigma}^2)$$

$$\tilde{\theta}_s \sim \text{Normal}(0, 100000)$$

Is our model specification complete?

Hierarchical model specification

- For $i = 1, 2, \dots, N$ observations, $c = 1, 2, \dots, C$ curves, plant $p = 1, 2, \dots, P$ plants, and $s = 1, 2, \dots, S$ species:



$$Pn_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = f(\alpha_{c(i)}, \beta_{c(i)}, Pmax_{c(i)}, Rd_{c(i)}, Q_i)$$

$$\theta = (\log(\alpha), \text{logit}(\beta), \log(Pmax), \log(Rd))$$

$$\theta_c \sim \text{Normal}(\hat{\theta}_{p(c)}, \hat{\sigma}^2)$$

$$\hat{\theta}_p \sim \text{Normal}(\tilde{\theta}_{s(p)}, \tilde{\sigma}^2)$$

$$\tilde{\theta}_s \sim \text{Normal}(0, 100000)$$

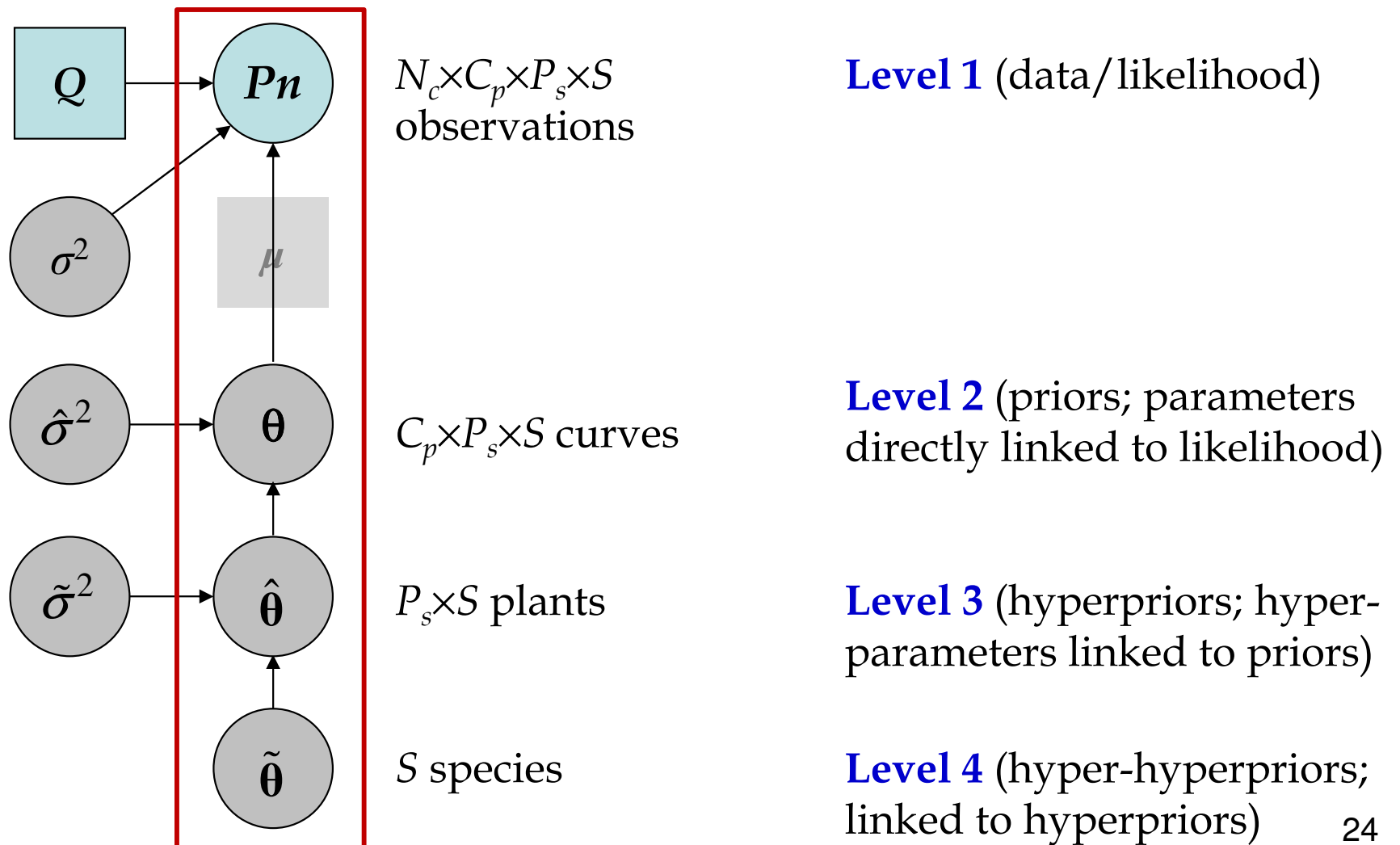
Common options for priors:

$$\tau = \sigma^{-2} \sim \text{Gamma}(0.01, 0.01)$$

$$\sigma \sim U(0, 100)$$

Hierarchical model w/ multiple levels

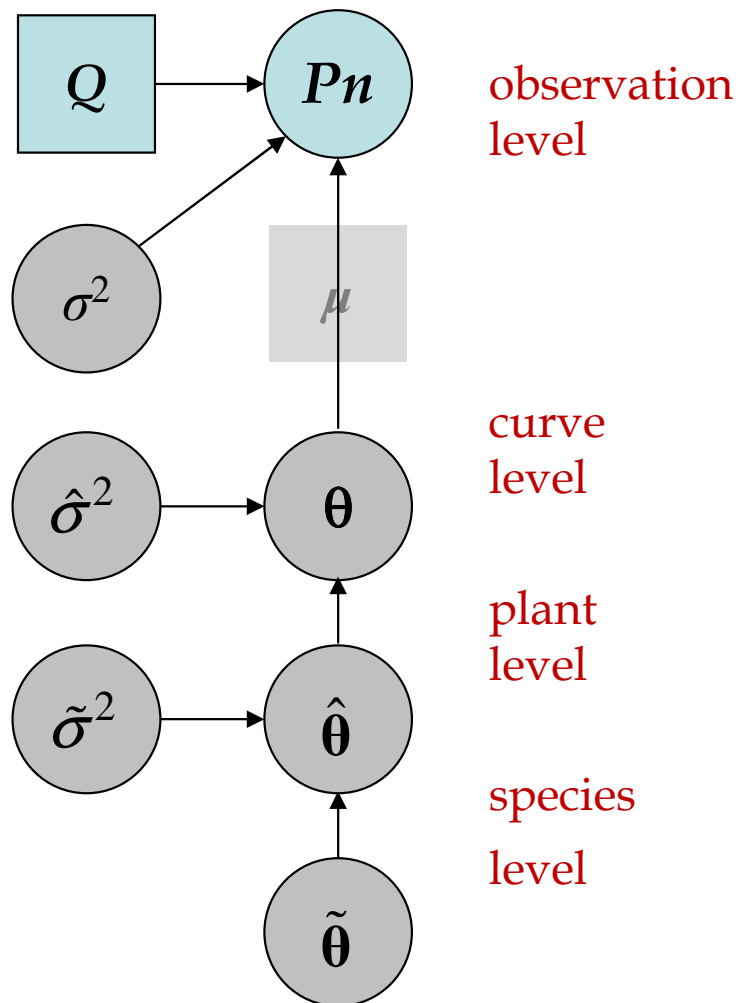
- Note dimension reduction when moving from level L to $L+1$
- Assuming full factorial design with N_c observations per curve (N total), C_p curves per plant (C total), P_s plants per species (P total), and S species



Write-out posterior

- **Goal:** obtain or sample from posterior, $[\theta, \hat{\theta}, \tilde{\theta}, \sigma^2, \hat{\sigma}^2, \tilde{\sigma}^2 \mid Pn, Q]$

$$[\theta, \hat{\theta}, \tilde{\theta}, \sigma^2, \hat{\sigma}^2, \tilde{\sigma}^2 \mid Pn, Q] \propto [Pn \mid \theta, \sigma^2, Q][\theta \mid \hat{\theta}, \hat{\sigma}^2][\hat{\theta} \mid \tilde{\theta}, \tilde{\sigma}^2][\tilde{\theta}][\sigma^2][\hat{\sigma}^2][\tilde{\sigma}^2]$$



Random vs fixed effects

- **Curves** and **plants** (*internal nodes*) are treated as *random effects* drawn from a population of effects.
- **Species** (root nodes) are treated as *fixed effects*.
- Comparatively large number of curves and plants; not interested in responses of individual curves or plants.
- Relatively few species; interested in species-level responses.

Implement model: Code

```
1 model{
2   for(i in 1:N){
3     # Likelihood of net photosynthesis (Pn) data:
4     Pn[i] ~ dnorm(mu[i], tau)
5
6     # Non-linear mean model; break-up into pieces for easier
7     # specification. Model parameters vary by curve ID:
8     part1[i] = alpha[curve[i]]*Q[i] + Pmax[curve[i]]
9     part2[i] = 4*alpha[curve[i]]*Q[i]*beta[curve[i]]*Pmax[curve[i]]
10    part3[i] = sqrt(pow(part1[i],2)) - part2[i]
11    AQcurve[i] = (part1[i] - part3[i])/(2*beta[curve[i]])
12    mu[i] = AQcurve[i] - Rd[curve[i]]
13  }
14
15  # Hierarchical priors for curve- and plant-level parameter.
16  for(c in 1:Ncurve){
17    # Grab appropriate component (element) of the theta parameter
18    # matrix and transform to original parameter scale:
19    alpha[c] = exp(theta[c,1])
20    beta[c] = ilogit(theta[c,2])
21    Pmax[c] = exp(theta[c,3])
22    Rd[c] = exp(theta[c,4])
23    # Define hierarchical priors on theta parameters, for each
24    # parameter type (k):
25    for(k in 1:4){
26      # Curves are nested in plant:
27      theta[c,k] ~ dnorm(theta.hat[plant[c],k], tau.hat[k])
28    }
29  }
30  # Hierarchical priors for plant-level parameters.
31  # Plants are nested in species:
```

Implement model: Code (cont.)

```
29 ^ }
30 # Hierarchical priors for plant-level parameters.
31 # Plants are nested in species:
32 v for(p in 1:Nplant){
33 v   for(k in 1:4){
34     theta.hat[p,k] ~ dnorm(theta.tilda[species[p],k],tau.tilda[k])
35 ^   }
36 ^ }
37 # Relatively non-informative priors for species-level parameters.
38 # Species treated as fixed effects (root nodes):
39 v for(s in 1:Nspecies){
40 v   for(k in 1:4){
41     theta.tilda[s,k] ~ dnorm(0,0.0001)
42 ^   }
43 ^ }
44
45 # Relatively non-informative, uniform priors for standard
46 # deviations in hierarchical priors. Compute corresponding
47 # precisions.
48 v for(k in 1:4){
49   sig.hat[k] ~ dunif(0,10)
50   sig.tilda[k] ~ dunif(0,10)
51   tau.hat[k] = pow(sig.hat[k],-2)
52   tau.tilda[k] = pow(sig.tilda[k],-2)
53 ^ }
54 # Conjugate, relatively non-informative gamma prior for precision
55 # in likelihood (data model):
56 tau ~ dgamma(0.1, 0.1)
57 # Compute associated standard deviation:
58 sig = 1/sqrt(tau)
59 ^ }
```

Implement model: Example data

- Curve-level observation data:

curve	Pn	Q
1	5.3738	1499.30
1	5.7832	997.60
1	5.9676	801.26
1	5.3067	498.30
...		
46	0.9463	97.89
46	0.1118	46.67
46	-1.8788	-0.28

- Samples sizes and indexing variables

N=445

Ncurve = 46

Nplant = 23

Nspecies = 4

plant = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 1, 3, 2, 1, 10, 11, 3, 2, 10, 1, 13, 14, 15, 16, 17, 18, 19, 20, 1, 11, 10, 3, 2, 13, 14, 12, 20, 15, 19, 16, 17, 18, 21, 22, 23)

species = c(1, 1, 1, 1, 1, 1, 2, 2, 2, 1, 1, 3, 3, 3, 4, 4, 4, 4, 4, 4, 1, 1, 1)

> length(plant)

[1] 46

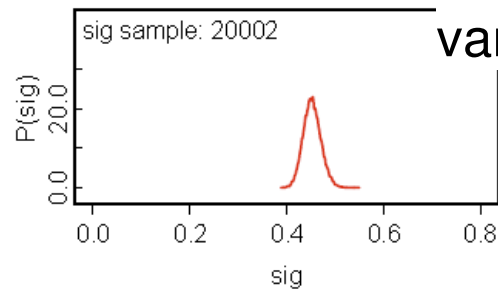
> length(species)

[1] 23

Example results / inferences

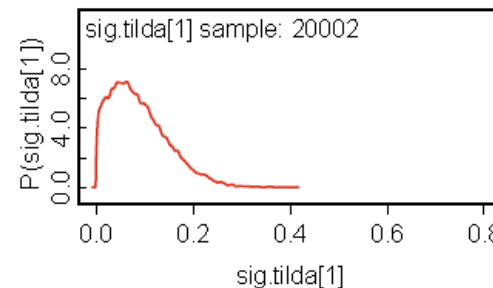
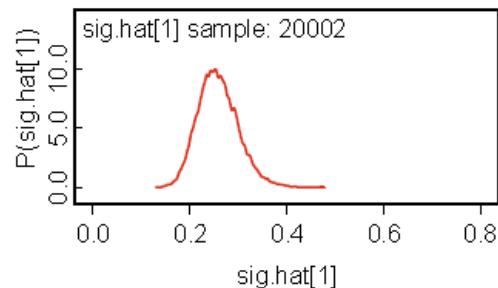
- Sources of uncertainty

σ



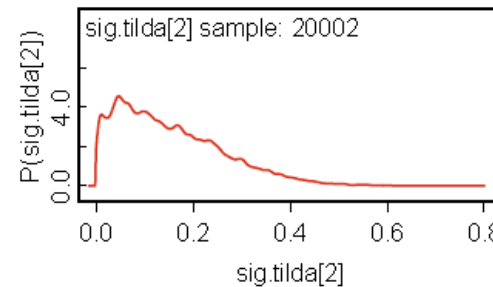
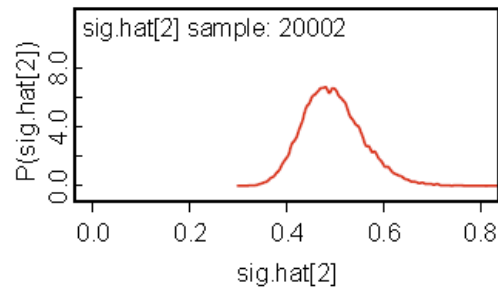
Observation
variability

$\hat{\sigma}_\alpha$



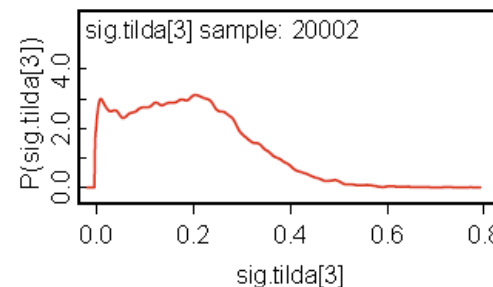
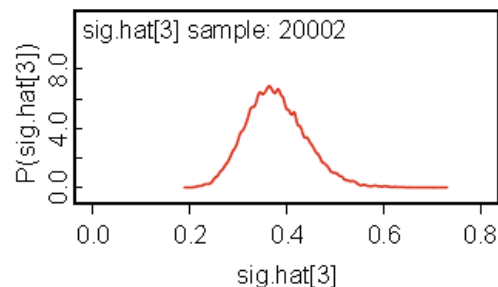
$\tilde{\sigma}_\alpha$

$\hat{\sigma}_{Pmax}$



$\tilde{\sigma}_{Pmax}$

$\hat{\sigma}_{Rd}$



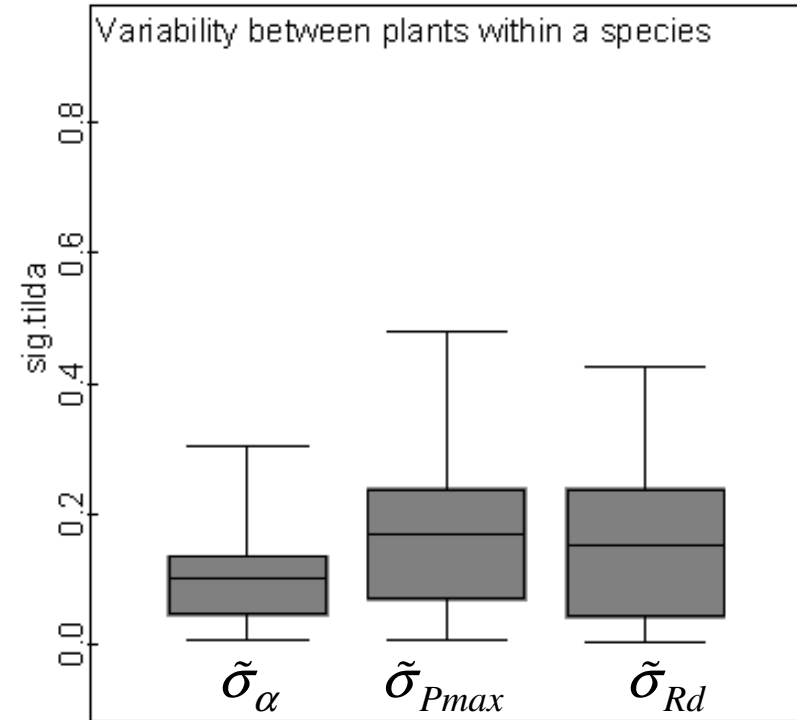
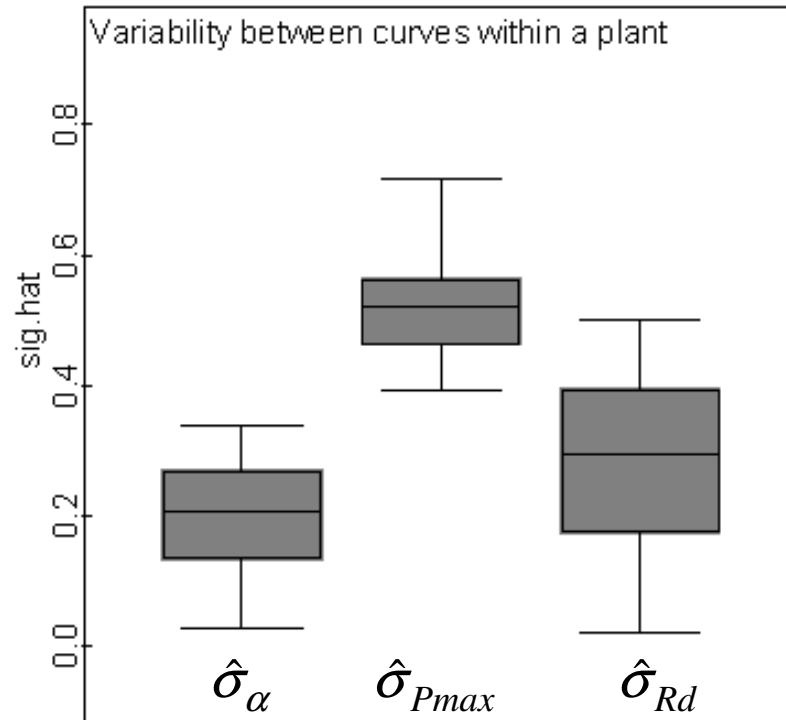
$\tilde{\sigma}_{Rd}$

Variability in
parameters
between
curves,
within a
plant

Variability in
parameters
between
plants,
within a
species

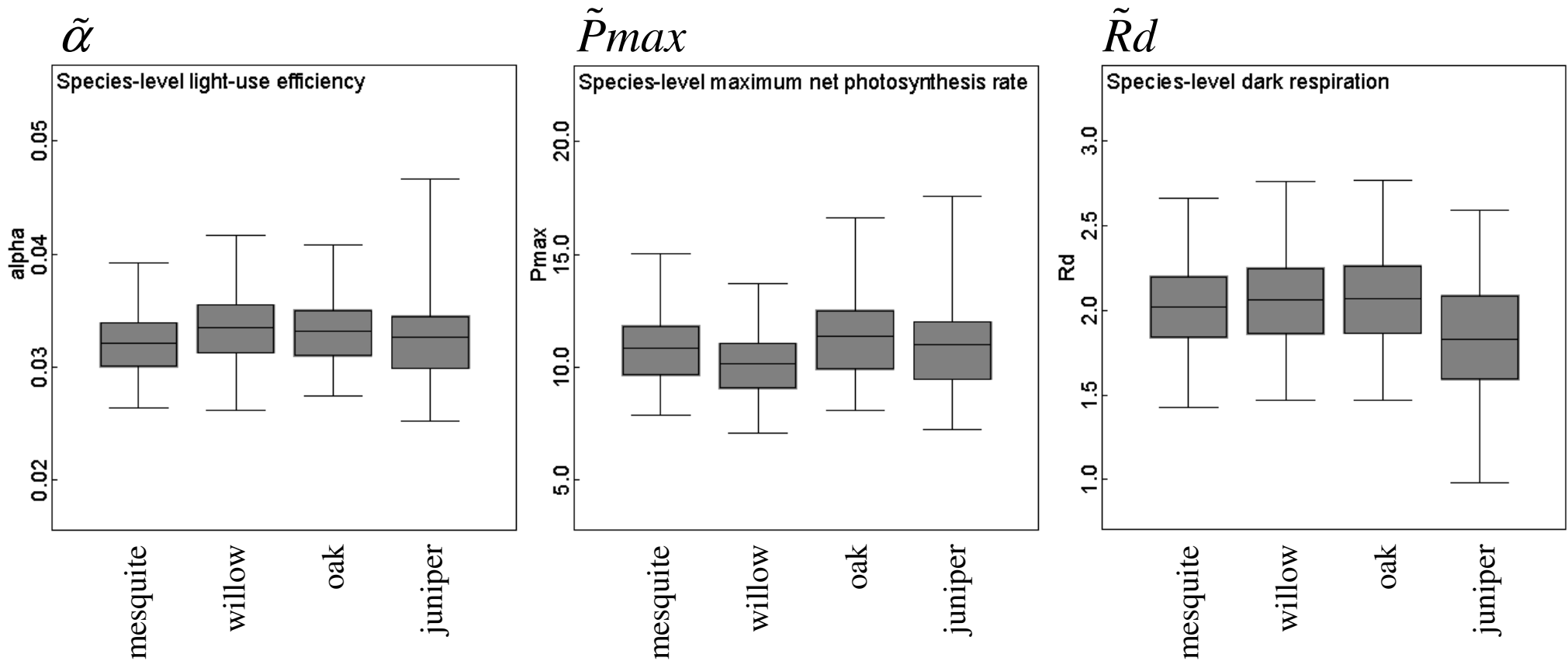
Example results / inferences

- Sources of uncertainty



Example results / inferences

- Species-level photosynthesis parameters



- See **Project 3** (light-response curves) for alternative mean model, different dataset, and alternative parameterization / model structure.

Appendix

- Alternative specifications of model
- Some approaches to parameterizing hierarchical models
- Advantages and disadvantages of different approaches

Alternative model specification

- Could choose a different *parameterization* that obeys support of original scale, for convenience or ease of interpretation
- Could use “moment matching” combined with distributions defined on the positive real-line to assign hierarchical priors to the curve-level α , β , $Pmax$, and Rd
- Instead, let θ represent a parameter on the *original* scale (i.e., α , $Pmax$, Rd) defined on positive real-line. We might assume $\theta \sim \text{Gamma}(a,b)$ or $\theta \sim \text{LogNormal}(m,v)$, e.g.:

$$\theta_c \sim \text{Gamma}(\hat{a}_{p(c)}, \hat{b}_{p(c)})$$

$$\hat{\theta}_p = E(\theta_c) \text{ and } \hat{\sigma}^2 = \text{Var}(\theta_c)$$

$$\hat{a}_p = \dots \quad \hat{b}_p = \dots$$

$$\hat{\theta}_p \sim \text{Gamma}(\tilde{a}_{s(p)}, \tilde{b}_{s(p)})$$

$$\tilde{\theta}_s = E(\hat{\theta}_p) \text{ and } \tilde{\sigma}^2 = \text{Var}(\hat{\theta}_p)$$

$$\tilde{a}_s = \dots \quad \tilde{b}_s = \dots$$

Assign priors to root node parameters:

$$\hat{\sigma}, \tilde{\sigma}, \tilde{\theta}_s$$

$$y \sim \text{Gamma}(a,b)$$

$$E(y) = \frac{a}{b} \quad \text{Var}(y) = \frac{a}{b^2}$$

$$a = \frac{[E(y)]^2}{\text{Var}(y)} \quad b = \frac{E(y)}{\text{Var}(y)}$$

What distribution would you choose for β ?

Recall, $0 < \beta < 1$

Some approaches to parameterizing hierarchical priors

- **Option 1:** Use hierarchical normal priors for parameters (e.g., θ), regardless of their support.
- **Option 2:** Transform θ to the real-line; e.g., if $\theta > 0$, use $\phi = \log(\theta)$; assign normal hierarchical priors to ϕ ; back-transform to obtain $\theta = \exp(\phi)$.
 - As in the light response curve example in this lecture
 - I do this frequently
- **Option 3:** Use prior that obeys support of θ (e.g., lognormal, gamma, etc. if $\theta > 0$; beta if $0 < \theta < 1$), assign hierarchical priors to $E(\theta)$, use moment matching to compute parameters (e.g., shape and scale) of the prior distributions.
 - Previous slide

Advantages/disadvantages of each approach

- **Option 1:** Use hierarchical normal priors for θ , regardless of support.
 - **Advantages:**
 - easy to specify/interpret
 - conjugates available at higher levels
 - usually works if the posterior for θ lies from away from unrealistic values
 - can implement “tricks” to improving mixing (e.g., parameter expansion to solve “zero variance traps” and “flat-lining”)
 - works well for hierarchical models with many levels
 - specification of starting values (usually) fairly straightforward
 - **Disadvantages:**
 - may cause errors if θ not defined on entire real-line
 - e.g., MCMC might sample a negative value when we know $\theta > 0$
 - this could result in numerical errors
 - requires careful specification of initial values

Advantages/disadvantages of each approach

- **Option 2:** Transform θ to the real-line; e.g., if $\theta > 0$, use $\phi = \log(\theta)$; assign normal hierarchical priors to ϕ ; back-transform to obtain $\theta = \exp(\phi)$.
 - **Advantages:**
 - conjugates available at higher levels
 - restricts θ to appropriate support
 - if assign prior to ϕ and compute $\theta = f(\phi)$
 - then modeling *median* hierarchically if f is monotonic
 - may be preferred if posterior for θ is highly skewed
 - can implement “tricks” to improving mixing (as in Option 1)
 - works well for hierarchical models with many levels
 - **Disadvantages:**
 - requires extra coding/calculations to compute θ
 - thus, may slow simulation time or result in errors (e.g., numerical overflow)
 - may want to model $E(\theta)$ hierarchically and not its median
 - specification of starting values more challenging

Advantages/disadvantages of each approach

- **Option 3:** Use prior that obeys support of θ (e.g., lognormal, gamma, etc. if $\theta > 0$; beta if $0 < \theta < 1$), assign hierarchical priors to $E(\theta)$, use moment matching to compute parameters (e.g., shape and scale) of the prior distributions.
 - **Advantages:**
 - restricts θ to appropriate support
 - allows you to model the mean (or any other moment) hierarchically
 - works best for hierarchical models with relatively few levels (as in the lab)
 - specification of starting values (usually) fairly straightforward
 - **Disadvantages:**
 - conjugates often not available at higher levels
 - requires extra coding/calculations to compute the distribution parameters defining θ 's prior
 - may slow simulation time or result in errors (e.g., numerical overflow)
 - can't easily use convergence/mixing “tricks” such as parameter expansion

Motivation for hierarchical models

- Flexibility and generality, can accommodate:
 - Unbalanced experimental designs
 - Missing data (*obtain posterior predictive distributions*)
 - Multiple data sources (*including prior sources of information*)
 - Non-linear processes (*example in this lecture*)
 - Non-Gaussian data or processes
 - Nested and non-nested effects
 - Spatial, temporal, individual effects
 - Multiple sources (or levels) of uncertainty
- Requires greater involvement of “user”
 - One must choose appropriate distributions
 - One must pick appropriate deterministic models
 - One must quantify underlying assumptions, processes, models
 - Good and bad...